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Assessing the Quality and Accuracy of Syphilis-Related Content on TikTok and YouTube: A Comprehensive Analysis

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Background: Sexually transmitted infections, including syphilis, pose a significant public health challenge. The advent of social media platforms has revolutionized health information dissemination, with YouTube and TikTok emerging as prominent sources. However, concerns persist regarding the reliability of syphilis-related content on these platforms. This study aimed to evaluate the quality and accuracy of syphilis-related content on TikTok and YouTube, using established tools such as DISCERN, Accuracy in Digital-Health Instrument (ANDI), and Global Quality Scale (GQS).

Methodology: We conducted a thorough search on TikTok and YouTube on November 26, 2023, using the keyword “syphilis.” Inclusion criteria comprised videos in English, less than 20 minutes in duration, and relevance to syphilis. Two dermatologists independently rated 98 eligible videos using DISCERN, ANDI, and GQS. Statistical analyses included χ^2 tests, mean comparisons, and interclass correlation.

Results: TikTok videos exhibited higher mean views ($222,519 \pm 412,746$) compared with YouTube ($127,527 \pm 223,622$). However, TikTok videos had lower mean GQS (2.3 ± 0.9), ANDI (2.19 ± 0.99), and DISCERN (28.7 ± 6.56) scores compared with YouTube (GQS: 2.9 ± 1.1 , ANDI: 2.90 ± 0.97 , DISCERN: 38.8 ± 9). Nonprofessional uploaders were 40.8% on TikTok, whereas the value was 53.1% on YouTube.

Conclusion: This study reveals disparities in the quality and accuracy of syphilis-related content on TikTok and YouTube. Despite higher popularity on TikTok, content quality, as assessed by DISCERN, ANDI, and GQS, was generally lower compared with YouTube. Targeted interventions are needed to improve the reliability of syphilis-related information on social media platforms.

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Conflict of Interest and Sources of Funding: None declared.

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Received for publication May 12, 2024, and accepted September 5, 2024. Supplemental digital content is available for this article. Direct URL citations appear in the printed text, and links to the digital files are provided in the HTML text of this article on the journal's Web site (<http://www.stdjournal.com>).

DOI: 10.1097/OLQ.0000000000002090

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BACKGROUND

Syphilis, a sexually transmitted infection caused by the bacterium *Treponema pallidum*, has been a persistent global public health concern throughout history.¹ First documented in the late 15th century during the Italian Renaissance, the disease has since traversed centuries, witnessing diverse epidemiological patterns, medical understanding, and societal perceptions.² The causative agent, *T. pallidum*, is a complex bacterium with the ability to invade multiple tissues and elude the host immune response, contributing to the chronic nature of the infection.¹ The transmission of syphilis occurs predominantly through sexual contact, presenting a significant challenge for prevention and control efforts.² The management of syphilis relies on effective antibiotic treatment, primarily penicillin, a testament to the enduring role of this antibiotic in combating a broad range of bacterial infections.³

The proliferation of social media platforms has revolutionized the way individuals' access, share, and engage with information, including health-related content.⁴ In recent years, platforms like TikTok and YouTube have emerged as influential channels for disseminating health information, offering a dynamic space where diverse voices, both professional and nonprofessional, contribute to the discourse.⁵ This evolution in information dissemination brings both opportunities and challenges, particularly in the transmission of public health information.^{5,6}

Social media platforms have become go-to sources for individuals seeking health information, owing to their accessibility, user-friendly interfaces, and the ability to deliver content in various formats, such as videos, images, and written text.⁷ The appeal lies in the immediacy and democratization of information sharing, enabling users to access insights and experiences beyond traditional health care channels.⁵ However, the open and unregulated nature of these platforms also raises concerns about the accuracy, quality, and reliability of the health information available.⁸

TikTok, with its short-form video format and emphasis on creativity and entertainment, has quickly become a global phenomenon, attracting a diverse user base.⁹ The platform's algorithm-driven content discovery ensures broad reach, making it an influential space for health-related content. However, the brevity and informal nature of TikTok videos pose challenges in conveying nuanced and accurate health information, especially on complex topics like syphilis.¹⁰

YouTube, on the other hand, has long been recognized as a hub for educational content, including health-related information.¹¹ With its diverse content creators, including health care professionals, medical groups, and individuals sharing personal experiences, YouTube offers a more extensive range of perspectives.¹² However, the sheer volume of content and varying levels of expertise among creators make it essential to scrutinize the quality and accuracy of health information on this platform.¹³

TABLE 1. Characters of Uploaded Videos on TikTok and YouTube

Parameter	YouTube	TikTok	P
Professional	23 (46.9%)	29 (59.2%)	0.225
Nonprofessional	26 (53.1%)	20 (40.8%)	
Nonprofessional			0.017*
Patient	0 (0%)	4 (20%)	
Not patient	26 (100%)	16 (80%)	
Verified	20 (40.8%)	5 (10%)	0.001*
Gender			0.327
No human in video	2 (4.1%)	4 (8.2%)	
Male	29 (59.2%)	22 (44.9%)	
Female	18 (36.7%)	23 (46.9%)	
Animation	23 (46.9%)	3 (6%)	<0.001*
Subtitle	34 (69.4%)	31 (63.3%)	0.521
Music	17 (34.7%)	22 (44.9%)	0.302
Written material	20 (40.8%)	1 (2%)	<0.001*
Educational	48 (98%)	44 (88%)	0.092
Anecdotal	0 (0%)	4 (8%)	0.041*
Home remedy	1 (2%)	0 (0%)	0.315
Advertisement	1 (2%)	4 (8.2%)	0.168

*Statistically significant, $P < 0.05$.

Given the dynamic nature of syphilis, the evolution of its epidemiology, and the changing landscape of health communication, it becomes crucial to assess the information available on these platforms. Understanding how syphilis is portrayed, discussed, and disseminated on social media can shed light on the knowledge gaps, misconceptions, and potential areas for public health intervention. This study aims to critically evaluate the accuracy and quality of syphilis-related content on YouTube and TikTok, using established tools such as DISCERN, Global Quality Scale (GQS), and Accuracy in Digital-health Instrument (ANDI), to assess the reliability of information and its potential impact on public health outcomes. By scrutinizing the characteristics of uploading channels, viewer engagement, and the presence of health care professionals, the study aims to provide insights into the quality, accuracy, and factors influencing health information dissemination on these popular social media platforms.

METHODOLOGY

Study Design

This research used a cross-sectional study design to systematically evaluate and analyze syphilis-related content on 2 major social media platforms, TikTok and YouTube. The study involves a comprehensive examination of the characteristics of uploaded videos, viewer engagement metrics, and the quality and accuracy of health information using established assessment tools.

TABLE 2. Video-Related Statistics (Mean $\pm$ SD)

Statistics	YouTube	TikTok	P
Duration, s	463.13 $\pm$ 938.94	57.18 $\pm$ 37.37	0.003*
Views	127,527.43 $\pm$ 223,621.56	222,518.57 $\pm$ 412,746.34	0.160
Daily views	340.15 $\pm$ 1099.61	2089.40 $\pm$ 6668.22	0.073
Likes	3874.92 $\pm$ 10,236.26	22,612.29 $\pm$ 79,487.55	0.05*
Comments	154.43 $\pm$ 334.25	1782 $\pm$ 6055.05	0.063
Engagement ratio	2.47 $\pm$ 3.39	44.31 $\pm$ 275.84	0.291
Followers	872,044.10 $\pm$ 1,830,388.81	950,780.92 $\pm$ 3,272,164.82	0.883

*Statistically significant, $P < 0.05$.

Sampling and Data Collection

The sampling process commenced from November 8 to 30, 2023, with a systematic search on TikTok using the incognito mode for the keyword “syphilis.” The inclusion criteria encompassed all relevant videos, and an additional 50 videos were added from one author’s account, sorted by views, resulting in a total of 100 TikTok videos. Similarly, on YouTube, a parallel process was followed, refining the search for “syphilis” and including the top 50 videos sorted by views. Duplicate videos were removed, and the final dataset included a total of 98 videos.

Video Selection Criteria

To ensure relevance and focus, videos not directly related to syphilis were excluded from the study. Furthermore, videos not in English and those exceeding a duration of 20 minutes were excluded.

Demographic Characteristics

The selected videos were meticulously analyzed to capture essential demographic characteristics, including the type of uploader (patient, health care professional, nonprofessional), professional designation for health care uploaders (physicians and health care organizations), and the categorization of uploading channels into distinct entities. Metrics reflecting viewer engagement were extracted, including the mean and SD of views, likes, shares, followers, and the presence of verification status.

Quality Assessment Tools

To evaluate the quality and accuracy of syphilis-related content, 3 established tools were used: DISCERN, GQS, and ANDI. DISCERN is a tool comprising 16 questions, measuring from 1 to 5, where 1 means “no” and 5 means “yes,” whereas 3 is “partial.” It measures 3 aspects, including reliability, quality of information, and overall rating of the videos. The GQS and ANDI’s primary aim is to provide a standardized method for assessing the overall quality of digital health content. All the tools have been used in many research studies.^{14–16} All tools used are provided in Appendix 1. Two independent expert dermatologists rated the videos, ensuring a comprehensive and professional assessment.

Statistical Analysis

Data analysis was performed using Statistical Package for the Social Sciences, SPSS 23rd version. Frequency and percentages were used to display categorical variables. Mean and SD were used to present numerical variables. χ^2 Test was used to test for the presence of association between categorical variables. Independent t test was used to test for the association between numerical variables and binary categorical variables. Level of significance was set at 0.05.

TABLE 3. Comparison of Video Assessment Using Different Tools Across Platforms

Item	Platform		<i>P</i>
	YouTube (n = 49)	TikTok (n = 49)	
Mean GQS	2.90 ± 1.13	2.31 ± 0.86	0.005*
Mean ANDI	2.90 ± 0.97	2.19 ± 0.99	0.001*
Mean DISCERN	38.80 ± 9	28.7 ± 6.56	<0.001*

*Statistically significant, *P* < 0.05.

As the GQS assessment, ANDI assessment, and DISCERN assessment were scored by 2 different evaluators, interrater reliability analysis was conducted to make sure a reliable scoring was carried out. The interrater reliability analysis was performed for YouTube videos and TikTok videos each on their own.

RESULTS

A total of 49 video from YouTube and 49 video from TikTok were included in the study. Table 1 shows the comparison of the uploaded video characteristics across the platforms, 46.9% of the videos in YouTube were uploaded by a professional, and 59.2% of the videos in TikTok were uploaded by a professional (no significant difference was observed between the platforms). A significant difference (*P* = 0.017) between YouTube and TikTok was observed in terms of having uploaded videos from patients from the nonprofessional uploaders (20% in TikTok vs. 0% in YouTube). As for the gender contribution in the uploaded videos, YouTube had higher male contribution (59.2%), whereas TikTok had higher female contribution (46.9%); however, there was no significant difference between the platforms. As for the verified accounts, YouTube-uploaded videos had a significantly higher rate of being uploaded by a verified accounts compared with TikTok (*P* = 0.001; 40.8% vs. 10.2%). YouTube videos also had a significant higher rate of having animation in the videos compared with TikTok (*P* < 0.001; 46.9% vs. 6.1%). Moreover, YouTube videos also had a significant higher rate of having written material in the videos compared with TikTok (*P* < 0.001; 40.8% vs. 6.1%). No significant difference between YouTube and TikTok was observed when comparing the presence of subtitle, music, education, home remedy, and advertisement. YouTube videos were significantly longer in duration compared with TikTok videos (*P* = 0.003; 463.13 ± 938.94 vs. 57.18 ± 37.37 seconds). TikTok videos had a significantly higher likes compared with YouTube (*P* = 0.05; 22,612.29 ± 79,487.55 vs. 3874.92 ± 10,236.26 likes). No significant difference was observed between YouTube and TikTok in terms of the following: views, daily views, comments,

engagement ratio, and followers; however, TikTok videos have higher views and daily views than YouTube videos (Table 2).

As for the interrater reliability analysis of YouTube videos, the interclass correlation for GQS scoring was 0.754 (indicating a strong agreement), the interclass correlation for ANDI scoring was 0.571 (indicating a moderate agreement), and the interclass correlation for DISCERN scoring was 0.599 (indicating a moderate agreement). As for the interrater reliability analysis of TikTok videos, the interclass correlation for GQS scoring was 0.599 (indicating a moderate agreement), the interclass correlation for ANDI scoring was 0.414 (indicating a moderate agreement), and the interclass correlation for DISCERN scoring was 0.564 (indicating a moderate agreement). When computing the interrater reliability analysis of YouTube and TikTok videos all together, the interclass correlation for GQS scoring was 0.718 (indicating a strong agreement), the interclass correlation for ANDI scoring was 0.532 (indicating a moderate agreement), and the interclass correlation for DISCERN scoring was 0.686 (indicating a moderate agreement).

Table 3 displays the comparison of video assessments using different tools across platforms. YouTube was found to have a significantly higher score compared with TikTok when comparing mean GQS (*P* = 0.005; 2.9 ± 1.13 vs. 2.31 ± 0.86), mean ANDI (*P* = 0.001; 2.9 ± 0.97 vs. 2.19 ± 0.99), and mean DISCERN (*P* < 0.001; 38.8 ± 9 vs. 28.7 ± 6.56).

Table 4 demonstrates the comparison of professional and nonprofessional content creators within YouTube. No significant difference was observed between professional and nonprofessional content creators in YouTube in terms of duration in seconds, views, daily views, likes, comments, engagement ratio, followers, mean GQS, mean ANDI, and mean DISCERN. Table 5 illustrates the comparison of professional and nonprofessional content creators within TikTok. It was observed that professional content creators in TikTok had a significantly higher assessment compared with the nonprofessional content creators in terms of mean GQS (*P* = 0.07; 2.59 ± 0.87 vs. 1.93 ± 0.69), mean ANDI (*P* < 0.001; 2.64 ± 0.85 vs. 1.55 ± 0.83), and mean DISCERN (*P* = 0.005; 30.84 ± 7.15 vs. 25.6 ± 4.03). No significant difference was observed between professional and nonprofessional content creators in TikTok in terms of duration in seconds, views, daily views, likes, comments, engagement ratio, and followers. Although not statistically significant, YouTube professional videos have higher views and similar engagement ratio in comparison to nonprofessional videos. On the other hand, TikTok has almost similar views but with higher engagement ratio for nonprofessional videos.

Figure 1, <http://links.lww.com/OLQ/B147>, available as Supplemental Digital Content, illustrates the comparison of video assessments across platforms using the DISCERN instrument. The results indicate a significant difference between YouTube and TikTok (*P* < 0.01), with YouTube videos showing a higher rate

TABLE 4. Comparison of Professional and Nonprofessional Content Creators Within YouTube

Factor	Content Creator in YouTube		<i>P</i>
	Professional (n = 23)	Nonprofessional (n = 26)	
Duration, s	665.46 ± 1328.17	284.15 ± 263.36	0.158
Views	154,767.7 ± 261,959.27	103,430.27 ± 185,305.6	0.428
Daily views	465.62 ± 1567.32	229.16 ± 364.09	0.458
Likes	5414.35 ± 13,497.39	2513.12 ± 6062.41	0.327
Comments	175.04 ± 361.93	136.19 ± 313.85	0.689
Engagement ratio	2.47 ± 3.56	2.47 ± 3.31	0.998
Followers	864,738.17 ± 1,841,305.79	878,507.04 ± 1,857,140.23	0.979
Mean GQS	3.02 ± 1.06	2.79 ± 1.2	0.477
Mean ANDI	3.13 ± 0.84	2.69 ± 1.05	0.117
Mean DISCERN	40.39 ± 9.62	37.38 ± 8.36	0.248

TABLE 5. Comparison of Professional and Nonprofessional Content Creators Within TikTok

Factor	Content Creator in TikTok		P
	Professional (n = 29)	Nonprofessional (n = 20)	
Duration, s	59.76 ± 36.07	53.45 ± 39.83	0.567
Views	226,371.69 ± 471,199.4	2,126,931.55 ± 321,132.42	0.938
Daily views	2969.26 ± 8563.51	813.61 ± 1169.35	0.271
Likes	14,467.62 ± 46,717.49	44,222.05 ± 110,426.67	0.201
Comments	867.17 ± 3340.48	3108.5 ± 8546.91	0.206
Engagement ratio	4.32 ± 7.277	102.3 ± 431.47	0.225
Followers	343,332.83 ± 480,861.17	1,831,580.65 ± 5,032,775.05	0.119
Mean GQS	2.59 ± 0.87	1.93 ± 0.69	0.007*
Mean ANDI	2.64 ± 0.85	1.55 ± 0.83	<0.001*
Mean DISCERN	30.84 ± 7.15	25.6 ± 4.03	0.005*

*Statistically significant, $P < 0.05$.

of “good” and “fair” quality, whereas TikTok videos had a greater proportion of “very poor” quality.

DISCUSSION

This study reveals critical insights into the quality and accuracy of syphilis-related content on social media platforms, specifically YouTube and TikTok. Our analysis, using established evaluation tools DISCERN, ANDI, and GQS, uncovered a notable disparity in content integrity. The findings indicate that, although social media can facilitate widespread access to health information, significant variations in content quality may influence public understanding and perceptions of syphilis. Furthermore, the evaluation metrics used in our study provide a framework for assessing the reliability of health-related videos, underscoring the importance of critical appraisal in the digital health landscape.

One noteworthy finding is the stark contrast in content quality between TikTok and YouTube. Despite TikTok videos enjoying higher mean views, likes, and shares, they consistently scored lower in GQS, ANDI, and DISCERN compared with their YouTube counterparts. This discrepancy suggests that, although TikTok may be a platform with immense potential for broad reach and engagement, it also poses challenges in maintaining the quality and accuracy of health-related information.¹⁷

Previous studies have consistently found disparities in the reliability and quality of health-related content across TikTok and YouTube platforms, with TikTok videos generally exhibiting lower reliability compared with YouTube videos.^{18,19} These findings suggest that YouTube videos are perceived as being of higher quality in terms of providing accurate and trustworthy health information. One possible explanation for this disparity could be attributed to the differences in content creation and moderation policies between the 2 platforms. YouTube, being an established platform with stricter content guidelines and moderation mechanisms, may attract more content creators who are committed to providing reliable health information. Conversely, TikTok's format and algorithmic-driven content recommendation system may prioritize entertainment value over factual accuracy, potentially leading to a higher prevalence of low-quality or misleading health-related content.

The lower content quality on TikTok aligns with concerns raised in previous literature about the platform's emphasis on brevity, entertainment, and user-generated content.²⁰ TikTok's format, characterized by short videos often created by nonprofessional individuals, might contribute to the limitations in conveying comprehensive and accurate health information. In contrast, YouTube, with its longer video format and a mix of content from diverse uploader types, appears to provide a more balanced landscape for health-related information.²¹

The role of uploader types emerged as a significant factor influencing the quality of syphilis-related content. Many TikTok videos uploaded by nonprofessional uploaders (40.8%), reflecting the platform's user-generated nature. This raises concerns about the accuracy and reliability of health information disseminated by individuals lacking professional medical expertise.²⁰

In contrast, YouTube exhibited a more diverse landscape of uploader types, with medical groups/foundations comprising 55.1%. This diversity likely contributes to a more balanced representation of perspectives and expertise on the platform. Professional uploaders, particularly those associated with medical groups and foundations, demonstrated higher content quality, reinforcing the importance of reliable sources in health communication.^{20,22,23}

These findings align with existing literature highlighting the significance of professional input in health-related content on social media.^{17–24} In a study conducted by Zhu et al.,⁹ the efficacy of health communication via TikTok was investigated by analyzing the content disseminated through TikTok accounts managed by Chinese provincial health committees. Their findings illustrated that TikTok effectively facilitates health communication by achieving widespread dissemination and presenting health-related information in an engaging manner. Both our research and prior studies emphasize the pivotal role of using social media platforms for health promotion initiatives and highlight the importance of using proficient communication strategies to effectively advance public health agendas within communities. In addition, our findings underscore the necessity of incentivizing verified health care professionals to actively contribute and engage in providing accurate and credible health information, particularly on platforms like TikTok where nonprofessional content prevails.^{22,24}

Verified accounts are often associated with credible and authoritative sources, contributing to user trust in the information presented.²⁰ Platforms should consider implementing measures to encourage and verify health care professionals, institutions, and authoritative health sources to enhance the overall quality of health-related content.²⁵ Users, in turn, can utilize verification status as a heuristic to identify reliable sources amid the vast sea of health information.

Recommendations

Our findings demonstrate the significant potential of TikTok as an influential platform for the dissemination of health-related information. We advocate for governmental agencies and public health care institutions to harness this social media platform as a means to effectively communicate public health messages by enhancing the production of high-quality content. Moreover, we advise patients to exercise caution when seeking information

about syphilis on TikTok, given the substantial variations in information quality across different sources. Future research should explore content quality on emerging platforms, consider a broader range of health topics, and investigate the effectiveness of interventions aimed at improving the accuracy of health-related information on social media.

Limitations

Our research had some limitations. First, it was a cross-sectional study that only analyzed a portion of the videos available on YouTube and TikTok. To obtain more accurate results, it would have been beneficial to include a larger number of videos. Nonetheless, it is important to note that most social media platform users tend to view only the first few pages of search results. Therefore, analyzing the first videos per keyword search and sorting method allowed us to gather a representative sample of the videos that are most likely to be seen during a user's search. Second, our study was limited by the fact that videos are a snapshot in time and are constantly changing. The videos we reviewed were downloaded on a specific date and time and may not reflect the most up-to-date or relevant information when users seek it. In addition, it is essential to underscore that our study was confined to the examination of YouTube and TikTok only, thereby not encompassing other prominent and widely used social media platforms where individuals may search for medical guidance.

CONCLUSIONS

In conclusion, our study contributes to the evolving discourse on health information dissemination on social media platforms. The disparities observed in content quality across TikTok and YouTube, along with the influence of uploader types, verification status, and the inclusion of written material, underscore the multifaceted challenges in ensuring accurate health information reaches the public. Addressing these challenges requires collaborative efforts from platforms, content creators, and health care professionals. Strategies should be implemented to incentivize verified health care professionals to contribute, promote the inclusion of written material, and encourage the presence of a human element in health-related videos. As social media continues to play a pivotal role in information dissemination, ongoing research and interventions are vital to safeguard public health by ensuring the accuracy and reliability of health-related content.

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Appendix 1 Tools.

TABLE A. DISCERN Instrument for Evaluating Health Information of Online Videos

Are the aims clear?	1–5 points
Does the video clip achieve its aims?	1–5 points
Is it relevant?	1–5 points
Is it clear what sources of information were used to compile the video?	1–5 points
Is it clear when the information used or reported in the video was produced?	1–5 points
Is the content balanced and unbiased?	1–5 points
Does the video provide details of additional sources of support and information?	1–5 points
Does it refer to areas of uncertainty?	1–5 points
Does it describe how each treatment works?	1–5 points
Does it describe the benefits of each treatment?	1–5 points
Does it describe the risks of each treatment?	1–5 points
Does it describe what would happen if no treatment was used?	1–5 points
Does it describe how the treatment choices affect overall quality of life?	1–5 points
Is it clear that there may be more than one possible treatment choice?	1–5 points
Does it provide support for shared decision making?	1–5 points
Overall rating of the video based on the answers to all of the above questions?	1–5 points

TABLE B. Global Quality Score (GQS) Used for Evaluation of Video Clips

poor quality; poor flow of the video; most information missing; not at all useful for patients	1
Generally poor quality and poor flow; some information listed, but many important topics missing; of very limited use to patients	2
Moderate quality; suboptimal flow; some important information adequately discussed, but other information poorly discussed; somewhat useful for patients	3
Good quality and generally good flow; most of the relevant information listed, but some topics not covered; useful for patients	4
Excellent quality and flow; very useful for patients	5

TABLE C. Accuracy in Digital-Health Instrument (ANDI)

Not accurate at all	0
Minimally accurate	1
Fairly accurate	2
Mostly accurate	3
Completely accurate	4